WHERE K9 ENDS

The cheapest entry passer over the full diploma distance, and the point at which it tears.

A diploma run over 256 streams of 1 terabyte each. Ten streams hold the full distance, two fall at 256 gigabytes. This paper describes the tear, how it was safeguarded, and what it means for the measured cost floor.

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What this is about

An earlier paper in this series measures how cheap a 64-bit hash finalizer can be and still pass the RRC ladder. The answer was **nine operation points**: a constant onto the input, then two 128-bit folds. Nothing cheaper passes, tested against 500 constant variants, hardware AES and an enumerated fragment of a near-champion family.

Passing there meant: 32 entry streams to 2 GB, and in the depth test 64 streams to 16 GB. What stayed open was the question the literature takes as its yardstick:

Does the cheapest passer hold the diploma too?

The diploma (RRC-64-40) pushes 256 streams of 1 terabyte each, 256 terabytes in total. When that run was finally taken to the end, on meer10, it cost a week of machine time with ten streams at once on eight physical cores. It is the test Evensen presented for NASAM.

What already existed, and what did not.

Martin Leitner-Ankerl measured mixers over **128** RRC streams to 2^40 with PractRand 0.95 (README January 2020), and a chain of this family is among them: `wyhash3_mix` shows no failure anywhere (2^40 in every one of his 128 cells, read off his tables on 30 August 2026). His test set, however, contains *no* complemented disguises and stops at 128 streams. This paper presents the same distance over **256** streams, including the complement half, and reports a failure. A related two-fold chain in his set, `mumxmumxx1`, spreads from 2^28 to 2^40 at a mean of 2^37, the same picture as here: individual disguises fall early, most hold. He collected the data without naming it as a finding.

The answer is a precise no.

The candidate holds ten streams over the full distance and falls on two others after 256 gigabytes. Neither failure is marginal: the probabilities admit no reading as chance, in the same test, at the same place in the output.

The candidate: lu9

The chain under test costs nine points and is called **lu9** in this work:

```
x ^= 0x9E3779B97F4A7C15;
x = mulfold(x, 0x781F94B96E8EDB3B);
return mulfold(x, 0xB853D68343F7525B);
```

mulfold is the 128-bit fold: multiply to 128 bits, XOR the two halves back down to 64. Cost 4 per fold, plus 1 for the leading constant = 9.

How it came about, and whom it belongs to.

It was not taken from the literature but constructed against a *measured* weakness: the bare double fold (cost 8) falls on counter streams because inputs of low multiplicative weight produce a thin 128-bit product — the fold has nothing to fold. A constant up front makes every input generic. Only afterwards did it become apparent that this exact pattern carries the wyhash family. An independent rediscovery, not an original design. The shape belongs to Wang Yi, not to this paper. The name **lu9** therefore denotes *this chain* with these constants and this measurement, not the shape.

What separates it from the original.

The candidate under test is not wyhash but its pattern with our own constants. So that the measurement says something about the original code as well, that code was measured separately (see below), taken verbatim from the source and checked bit for bit against an independent reference.

The finding

The diploma run tests the most dangerous disguises first: all the streams without complement (where the known fold weakness lives), and within those the historically deadliest rotations up front. After twelve completed streams the verdict stood.

Streams	Result
10 streams	clean to 2^40 — the full terabyte, no finding
T0C0r40	**FAIL at 2^38** (256 GB), test [Low1/64] FPF-14+6/16, p = 6.8×10^-19
T1C0r02	**FAIL at 2^38** (256 GB), test [Low8/64] FPF-14+6/16, p = 3.0×10^-27

Why this is not a one-off artefact.

Two *independent* streams (different transformation, different rotation) fall at the same amount of data, in the same test type, at the same place in the output: the lowest bits. The probabilities are beyond 10^-18; PractRand reports over 900 further sub-tests with nothing to note alongside them. None of that reads as noise. It is a structure.

Why the lowest bits.

In a multiplication the lowest output bit depends only on the lowest input bit, the second on the lowest two, and so on: carries only travel upwards. The fold (upper half XOR lower) repairs that partially but not completely. On counter streams, where the low bits carry the actual structure, enough remains to demonstrate it after 256 gigabytes. Whether a stream falls depends on which rotation moves this weakness into a position the test can see. That is why the test measures all 256.

The safeguards

A tear at 256 gigabytes is exactly the kind of claim where you suspect your own instrument first. Three checks belong to it:

1. Can the rig judge at this depth at all?

The actual test: a mixer that per the literature passes a stage DEEPER (NASAM, RRC-64-42) runs on exactly the same two streams to 1 terabyte. If it holds, the rig measures correctly there. If it falls too, the problem is the rig. The depth anchor held: NASAM ran on the same two streams to 1 TB with no finding. The measuring path judges correctly at that depth, so the tear belongs to the candidate and not to the instrument.

2. Can the rig reject at all?

Carried in the same run: a known weak mixer (Stafford mix13) on the same stream. It fell as expected at 2^19; the positive control is in place.

3. Does the tear hang on the kind of constant?

Also carried along: the same chain with **addition instead of XOR** for the leading constant, otherwise identical. It falls as well, on the same stream, at **2^39**: one stage later, but it falls. That answers what this run was for: **the weakness hangs on the pattern (two folds) and not on the kind of first step.** The leading constant moves the tear; it does not remove it.

What needs no further check.

Reproducibility: the stream is deterministic. The same chain through the same disguise yields byte for byte the same sequence, so a repeat run necessarily produces the same verdict, and there is no spread to report because nothing is drawn at random. The counter-check is that the tear appears on two *different* disguises, T0C0r40 and T1C0r02. Different, not statistically independent: both are deterministic images of the same counter, and a mixer with one weakness can show it in many disguises at once. Ten of the twelve streams held the full terabyte, which is what makes the two informative rather than a repeat of one.

The original code on the same ladder

Because the candidate under test is only the *pattern*, the real code from the source was measured, taken verbatim and verified bit for bit against an independent reference implementation (65,536 values per variant, zero deviations).

What	Cost	On the ladder
The generator's mixing step, read as a finalizer for counter inputs	5	**FAIL at 2^13–2^14** — after 8 to 16 kilobytes
The hash path for an 8-byte word	~10	holds all four gauntlet streams to 2^28; depth test outstanding

One precision, stated explicitly.

The first line is **not** a statement about the generator. Natively it runs on a counter with a large stride; the measurement here pushes a counter with stride 1 through the same mixing step. The correct sentence is: *used as a finalizer for structured counter inputs, this mixing step dies after 8 kilobytes.* Turning that into “the generator is broken” claims something other than what was measured here.

The value of this measurement lies elsewhere: it confirms the cost floor with shipped production code. A five-point mixer from practice dies on the same streams on which thousands of search candidates have died.

What follows from this

The floor stands. The question splits.

Nothing in this finding refutes the measured floor: still nothing below nine points passes the entry test. What changes is the reach of the statement. One question becomes two, and since 27 August both have an answer:

Question	Answer
What does the **entry test** cost (32 streams at 2 GB)?	**9** — measured, holds against 500 constants, hardware AES, a partly enumerated family (1,309 of 30,980) and shipped production code
What does the **diploma** cost (256 streams at 1 TB)?	**10** — measured as an upper bound: meer10 passes 256/256. Nine tears at 2^38.

The region that was unmeasured.

The gap between 10 and 15 was the most interesting region of the space, and nothing was known about it. **meer10** (fold, double xor-shift, fold) was named here as the next candidate, on the strength of the depth test (64 of 64) and of not appearing anywhere in the literature check. It has since been run: 256 streams at 1 terabyte, **not one FAIL**.

So the two heights sit close together. lu9 and meer10 share a skeleton of three operations and differ in the middle one: a constant against a double shift-xor. Between surviving 64 gigabytes of disguised counter and surviving a quarter of a million lies that exchange, which is one cost point in the weights used here and two under the alternative the README sets out. Either way it is the same kind of step that separates K8 from K9: not more mixing, but mixing in the right place.

What the 10 does and does not say.

It is an upper bound, established by construction: a chain of cost 10 passes, so 10 suffices. It is *not* a proof that nine cannot. What is measured is that the two K9 chains taken to diploma distance both fell, and that of the K9 chains passing the entry test none but the wyhash pattern survived even the depth test. Longer nine-point chains remain unmeasured.

A note on the evidence for those 256 streams.

Four of the 256 raw logs were lost to an overwrite by the rig's own anchor and were measured again from scratch on 28 August: all four clean to 2^40, identical to what the checkpoint had recorded seven days earlier. All 256 are complete on disk. The incident is treated in *The Calibration Weights*.

Placing this in the literature.

Evensen himself tested murmur3, splitmix64, Moremur, NASAM and others over RRC, but no fold construction. Ankerl (2020) measured fold chains, without complement and at half the stream count. Vigna found a tear in the bit-reversed output of the generator after 32 terabytes in 2023, acknowledged in the project README. That is considerably later than the one measured here, but on a different question: generator output versus a bare finalizer on RRC streams. Three data points, the same direction.

What this is — and what it is NOT

It IS: a fully logged diploma measurement.

Every stream is booked individually, with verdict and runtime; the rig is documented openly, the verdict depends only on the byte stream and can be reproduced on any machine.

It is NOT a statement about wyhash as a hash.

What was measured is a 64-bit finalizer on structured counter streams. How the complete hash behaves on real keys is a different question, answered by different tools (SMHasher and relatives). Anyone reading this measurement as “wyhash is insecure” is reading it wrong.

It is not a proof of an upper bound.

That this chain tears at 2^38 says nothing about whether some other nine-point chain might make it after all. Of the 30,980 nine-point three-op chains, 1,309 were decided (about a quarter of one of three shape families) before a cap stopped the run, and not one of them survived the depth stage. Families B and C were never started; longer nine-point chains are unmeasured. The enumeration bounds nothing. It says only that where it looked, it found nothing.

The working state is marked.

The diploma run was stopped as planned after the second failure, since a further 244 streams would not have changed the verdict and would have cost 244 terabytes. So what is missing is known: it is not measured how many of the 256 streams would have fallen in total. The depth anchor is in.

In brief

The test	RRC-64-40: 256 streams at 1 TB, most dangerous first, stop at the first failure.
The result	10 streams clean to 1 TB, 2 streams FAIL at 256 GB (p = 7e-19 and 3e-27), both on the lowest bits.
The mechanism	The low bits of a multiplication depend on too few input bits; the fold does not fully heal that. The variant with addition instead of XOR falls at 2^39 — the tear sits in the pattern, not in the constant.
The context	Shipped production code of the same family: the five-point mixing step dies at 8 KB, the ten-point hash path holds the gauntlets.
The status	Cost floor for the entry test unchanged at 9. The diploma bound is measured at 10: meer10 passes 256/256 — the candidate this paper named for that region.

The name lu9 is not an abbreviation.
For L.

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As of: 28 August 2026 (working draft). Measuring rig: see *The Ladder*. Cost model and floor: see *The*

Cost Limit. The enumerated near-family: see *The Specious Family*. What one more cost point buys, measured over the same distance: see *MEER10*.